

Sample machine learning engineer job description

[Company X] is looking for a machine learning engineer to build efficient, data-driven artificial intelligence systems that advance our predictive automation capabilities. The candidate should be highly skilled in statistics and programming, with the ability to confidently assess, analyze, and organize large amounts of data. The candidate should also be able to execute tests and optimize [Company X]'s machine learning models and algorithms.

Objectives of this role

- Design and develop machine learning algorithms and deep learning applications and systems for [Company X]
- Solve complex problems with multilayered data sets, and optimize existing machine learning libraries and frameworks
- Collaborate with data scientists, administrators, data analysts, data engineers, and data architects on production systems and applications
- Identify differences in data distribution that could potentially affect model performance in real-world applications
- Ensure algorithms generate accurate user recommendations
- Stay up to date with developments in the machine learning industry

Responsibilities

- Study and transform data science prototypes and apply appropriate machine learning algorithms and tools
- Run machine learning tests and experiments, and document findings and results
- Train, retrain, and monitor machine learning systems and models as needed
- Construct optimized data pipelines to feed machine learning models
- Consult with managers to determine and refine machine learning objectives
- Extend existing machine learning libraries and frameworks

Skills and qualifications

- Impeccable analytical and problem-solving skills
- Extensive math and computer skills, with a deep understanding of probability, statistics, and algorithms
- In-depth knowledge of machine learning frameworks, like Keras or PyTorch

- Familiarity with data structures, data modeling, and software architecture
- Excellent time management and organizational skills
- Desire to learn

Preferred qualifications

- Proven experience as a machine learning engineer or similar role
- Familiarity with Python, Java, and R
- Excellent communication and collaboration skills
- Innovative mind with a passion for continuous learning
- General knowledge of building machine learning systems
- Bachelor's degree (or equivalent) in computer science, mathematics, or related field